Week 4 — Bayesian Generalization

Friday, May 22, 2026

Prof. Joseph Austerweil

Agenda

Welcome + Clusters walkthrough	0:00
The generalization problem	0:10
The Bayesian generalization framework + size principle	0:18
Rectangle game + number game	0:40
Break	1:08
Student presentation — Shohei	1:15
No Free Lunch	1:40
Hierarchical Bayes + close	1:50

Assignment 1 — Clusters

Clusters — what and when

Assignment 1: Gaussians, Categories, and Clusters

- Due **Fri Jun 5, 2026, 8:00 PM** — worth **7.5%**
- **Problem 1** — Gaussian-Gaussian conjugate model → *you can do this today* (it's Week 3 material)
- **Problem 2** — Gaussian mixture / categorization → Bayes' rule over two category Gaussians
- **Problem 3** — clustering → leans on this week + T3 Ch 5

Clusters — which notebook

- `clusters.ipynb` **is the canonical stencil** — the GenJAX path
- `clusters_python.ipynb` and `clusters_nosoln.Rmd` — non-GenJAX paths, *same math, same credit*
- Matlab available on request
- **"Open in Colab"** links are live on the assignments page — one click, no local setup
- Read the **assignment PDF first** — it has the problem statements and all the math

Assignments page: hml.chibatech.dev/assignments.html

The generalization problem

Chibany’s lunches

Chibany has had three tonkatsu lunches this week. Each weighed about **500 g**.

Today a lunch arrives weighing **480 g**. Is it tonkatsu?

What about **700 g**? What about **350 g**?

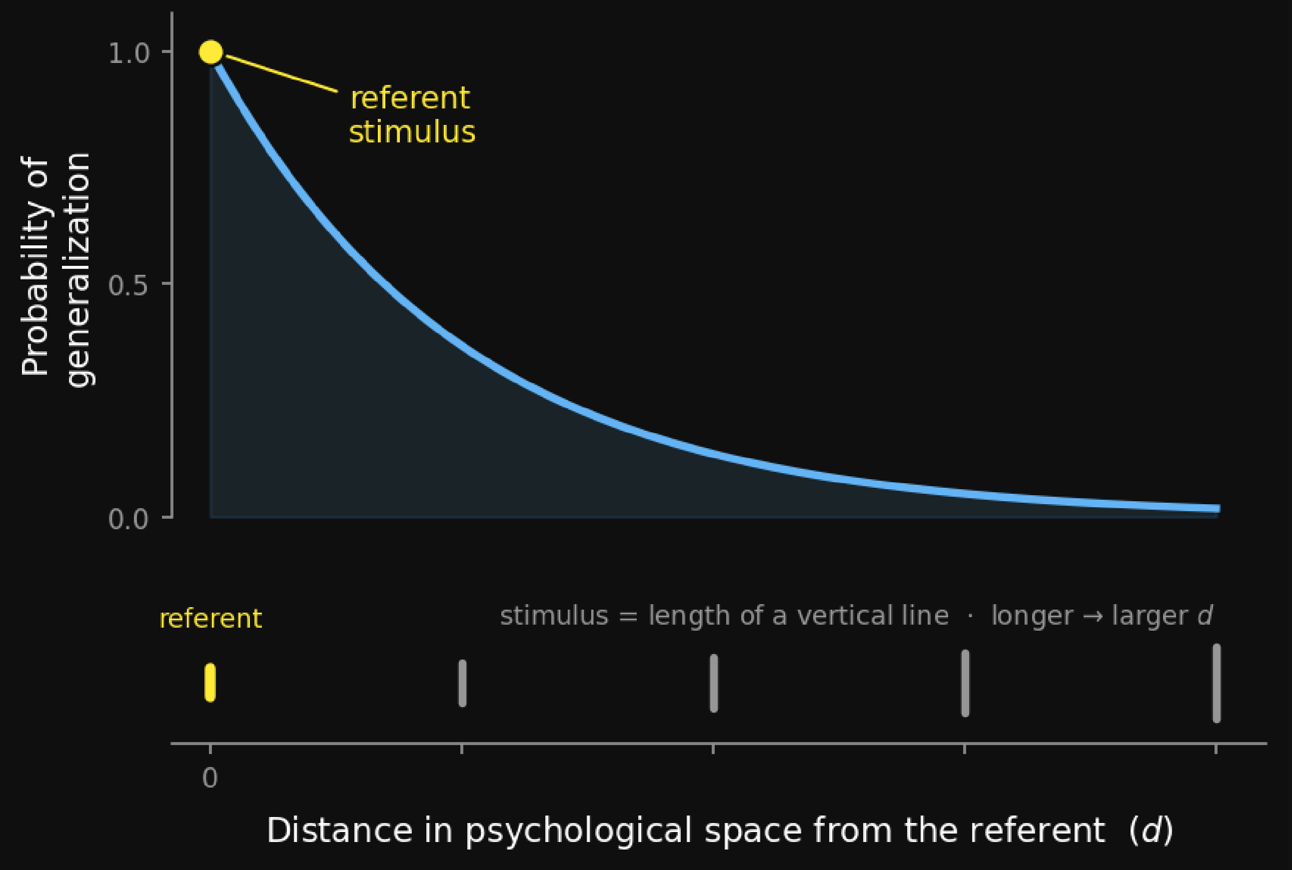
What just happened

Generalization — deciding when to extend a property from observed examples to a *novel* stimulus.

- No two stimuli are ever identical → you *must* generalize to act at all
- It is everywhere: word learning, categorization, object recognition, property induction, stereotypes
- It is the core problem of inductive inference

Shepard's universal law

Shepard (1987) — across species and domains, the probability of generalization **decays exponentially** with distance in **psychological space** (*the perceived distance, after the mind represents the stimulus — not raw physical space*).



One law, one equation:

$$g(d) = e^{-d}$$

- g — probability of generalization
- d — distance in psychological space from the referent

The same curve holds across **species, senses, and stimulus types** — pitch, colour, line length, faces. Shepard called it *universal* for that reason.

But the law is **descriptive**: it says generalization decays exponentially, not *why*. Today's Bayesian framework will **derive** that exponential — it falls out of the posterior.

Poll — Shepard’s universal law

Shepard’s universal law of generalization says generalization …

- **A.** decays exponentially in psychological space
- **B.** decays exponentially in stimulus space
- **C.** grows exponentially in psychological space
- **D.** grows exponentially in stimulus space

Poll — answer

A. Decays exponentially in psychological space.

“Stimulus space” is the trap — generalization isn’t governed by physical distance but by *perceived* distance. A model of generalization therefore needs a model of the **psychological space**. That is exactly what the Bayesian framework supplies next.

The Bayesian generalization framework

The idea — concepts as hypotheses

Instead of measuring distance directly, posit a **space of candidate concepts** and let Bayes do the generalizing.

- A concept = a **set of stimuli** that share the property
- Shepard's term: a **consequential subset** — the subset of things the property "applies to"
- Generalization becomes: *which concepts are consistent with the examples, and do they contain the new stimulus?*

Notation lock-in

- h — a **hypothesis**: one candidate concept, i.e. a *set* of stimuli
- $\mathcal{H}$ — the **hypothesis space**: all candidate h
- $X = \{x_1, \ldots, x_n\}$ — the observed **examples** of the concept
- y — a **novel stimulus** we must judge
- C — the (unknown) true concept

The three ingredients

Prior $p(h)$ — domain knowledge: which concepts are *natural* before any data.

Likelihood $p(X \mid h)$ — how probable the examples are if h is the true concept.

Posterior $p(h \mid X)$ — belief in h after seeing the examples:

$$p(h \mid X) \; \propto \; p(X \mid h)\, p(h)$$

The hypothesis space IS a prior

Choosing $\mathcal{H}$ is already a **strong prior**.

Any concept not in $\mathcal{H}$ has $p(h) = 0$ — it can never be learned, no matter the data.

So: a learner's inductive bias lives in *which hypotheses it even considers*.

Generalization = a posterior-weighted vote

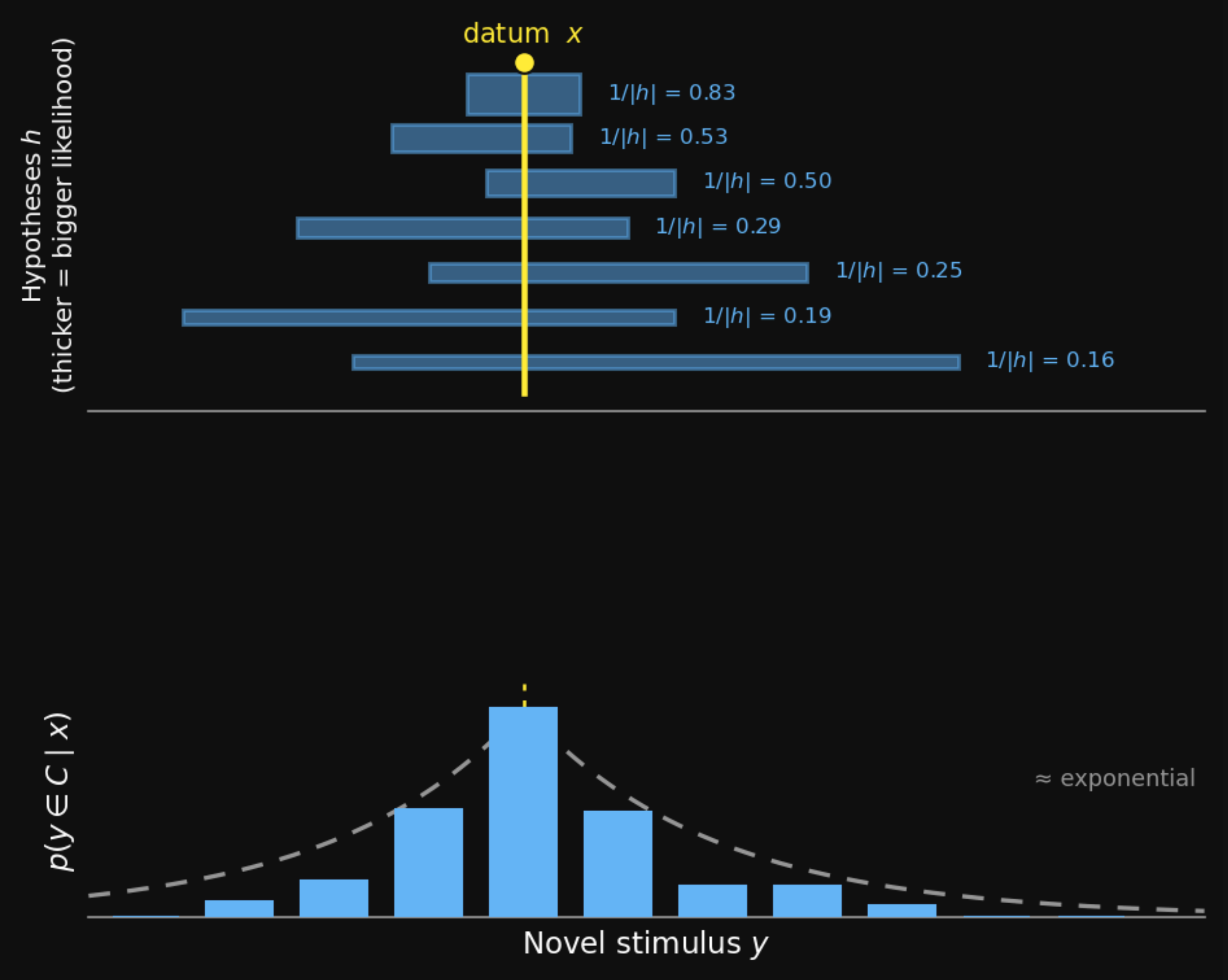
Probability that the novel stimulus y has the property:

$$p(y \in C \mid X) \;=\; \sum_{h \in \mathcal{H}} \mathbf{1}[\,y \in h\,]\, p(h \mid X)$$

Every hypothesis votes. Its vote is **its posterior weight**, and it votes “yes” only if it *contains* y.

$\mathbf{1}[\,y \in h\,]$ — the indicator: 1 if y is in the set h, else 0.

One datum → the posterior-weighted vote



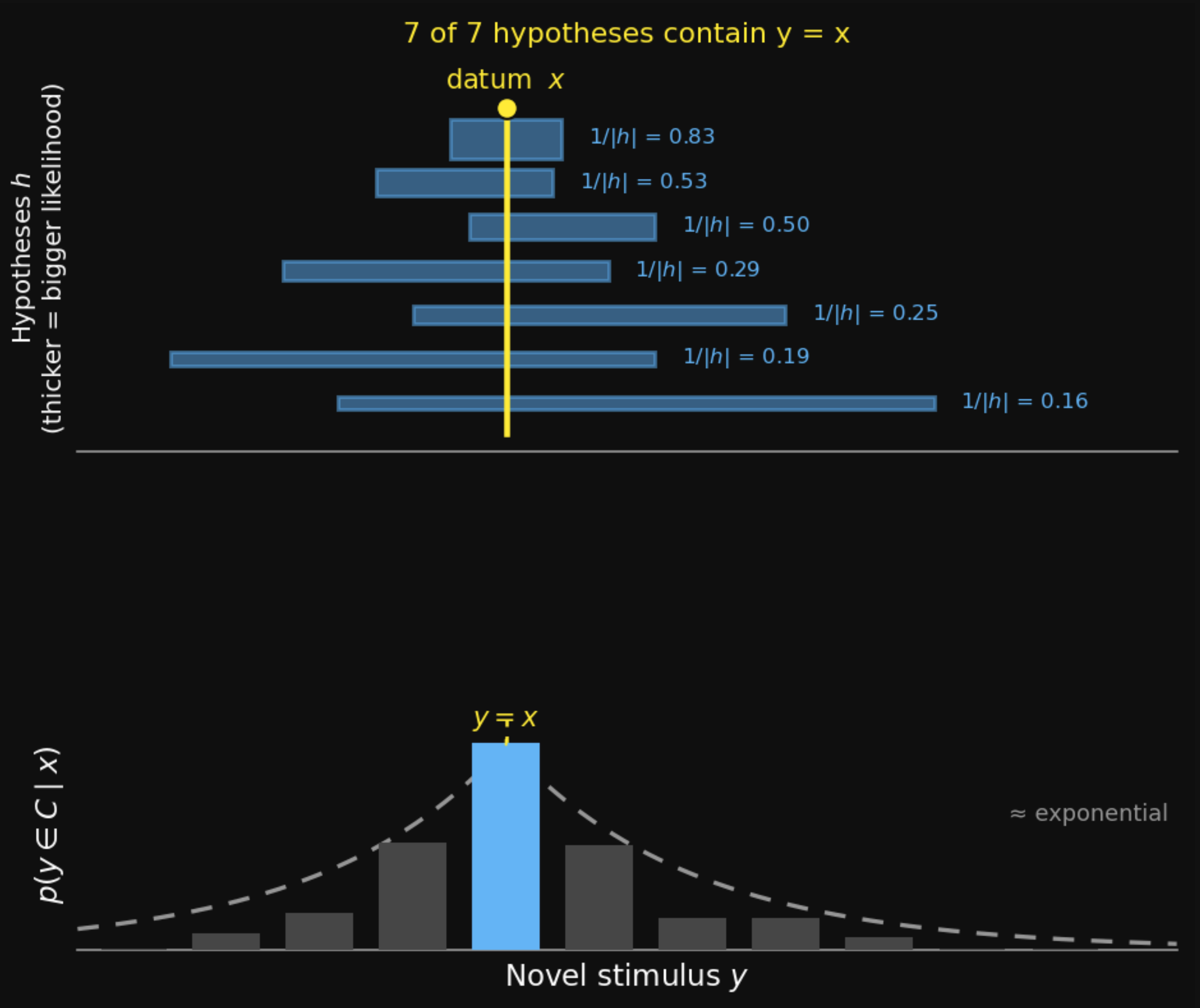
The hypotheses. Observe one datum x. Every interval that **contains** x is a live hypothesis h. A bar's thickness is its strong-sampling likelihood $1/|h|$ — a *smaller* interval is a *thicker* bar. Flat prior, so each h's posterior weight $\propto$ its likelihood.

The vote. For each candidate y, add up the posterior of every h that contains it:

$$p(y \in C \mid x) = \sum_h \mathbf{1}[\, y \in h \,]\, p(h \mid x)$$

Next three slides: walk y outward — x, $x{+}1$, $x{+}2$ — and watch the vote shrink as hypotheses drop out.

Vote for $y = x$



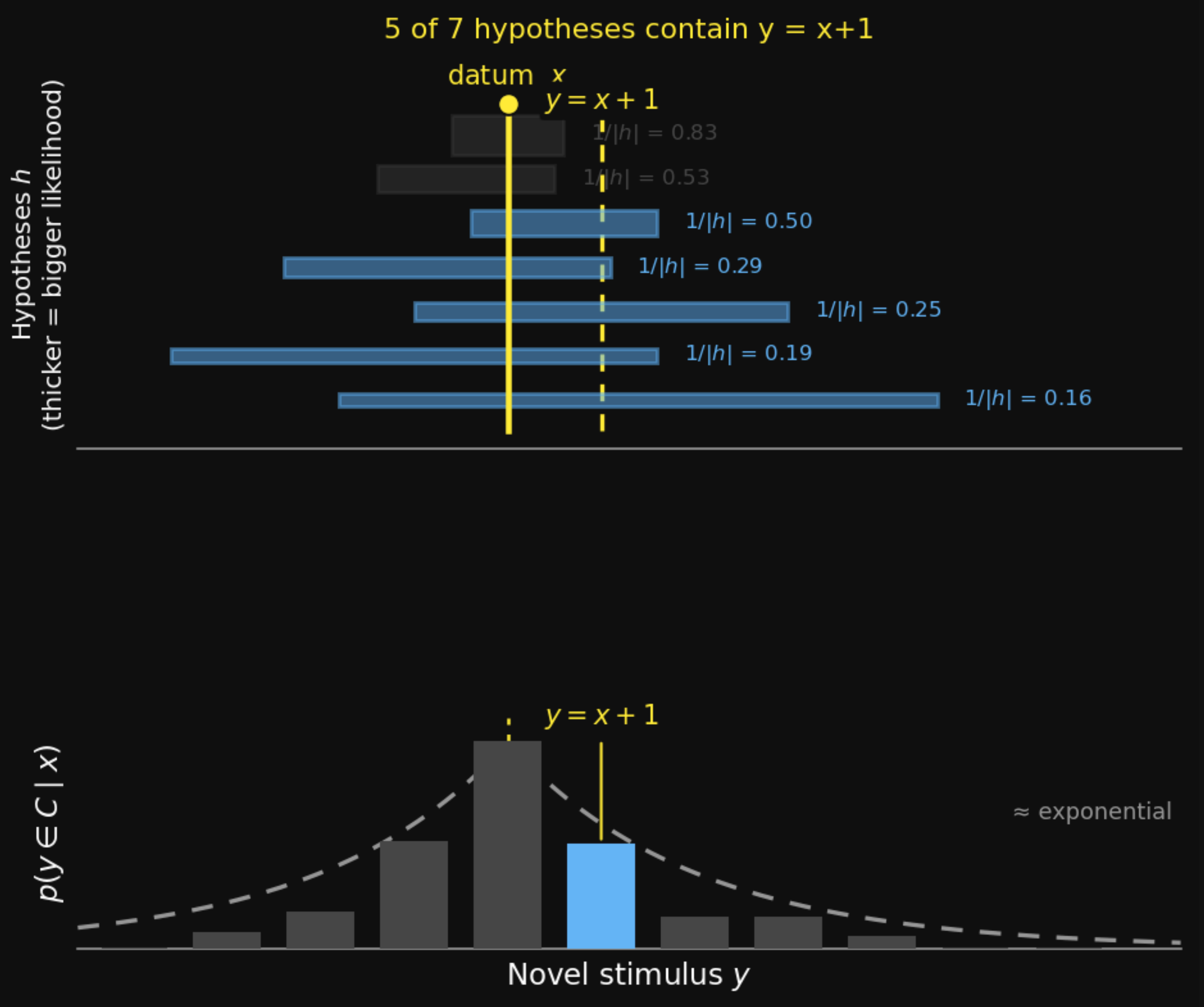
The question. Does the property generalize to $y = x$ — the observed datum itself?

Who votes. Every hypothesis in $\mathcal{H}$ was built to contain x, so **all 7 of 7 vote "yes"**. Not a single one drops out — there is no hypothesis that excludes the very point you observed.

The bar. The vote at x sums *all* the posterior weight — $\sum_h p(h \mid x) = 1$. This is the **tallest bar the gradient can have**.

The baseline. Generalization to the datum itself is near-certain — and it is the peak that every other y is measured against. As we step y away from x, hypotheses start dropping out, and the bar can only fall.

Vote for $y = x + 1$



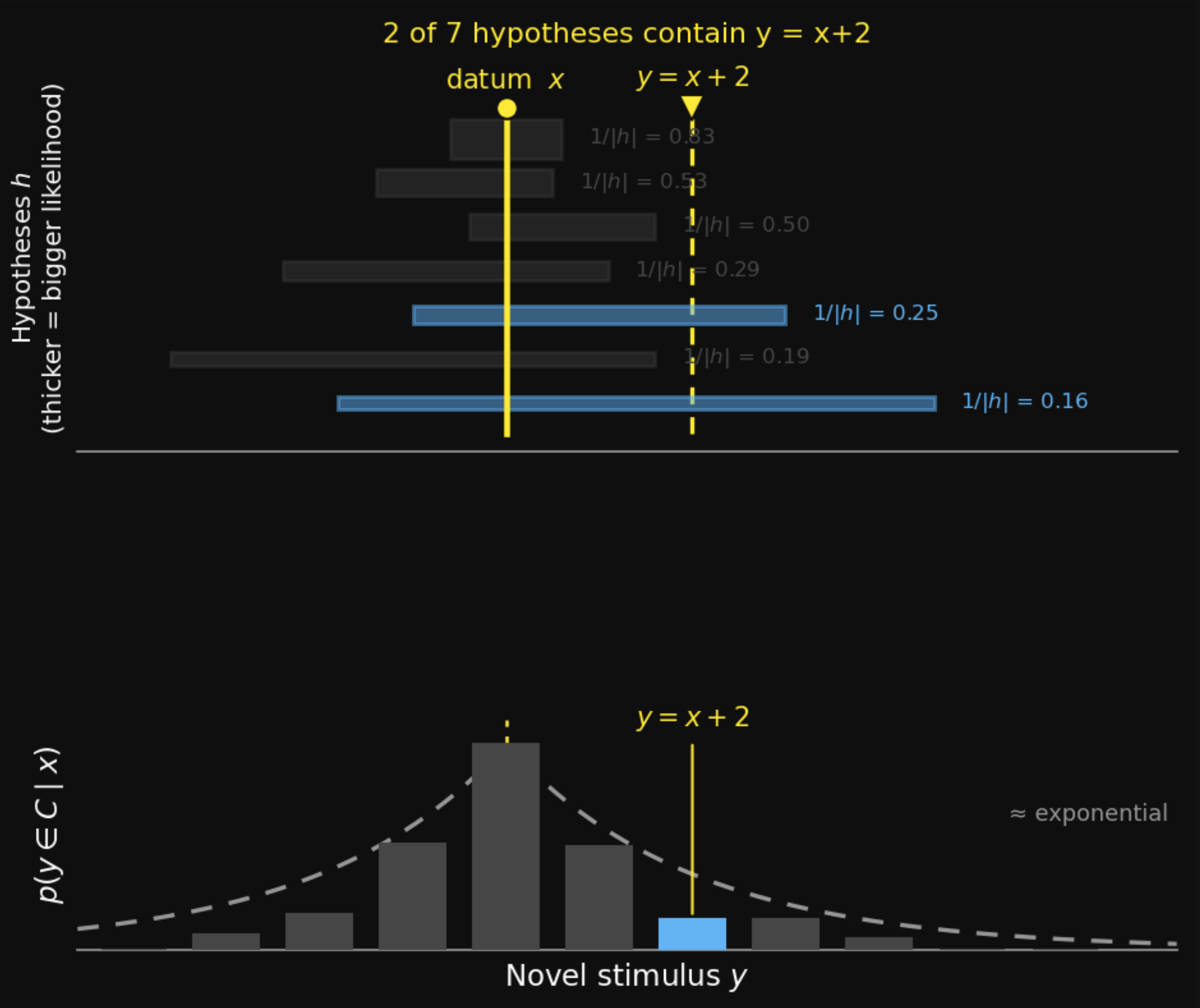
The question. Step one unit out from the datum: does the property generalize to $y = x + 1$?

Who drops. The two smallest intervals no longer reach y — y falls outside them, so they grey out and vote “no”. **5 of 7 still vote “yes”.**

The bar. Fewer hypotheses contribute → a **shorter** bar than at x.

Why it falls fast. Note *which* hypotheses dropped: the **smallest** intervals — and by the size principle those carry the **most** posterior weight (thickest bars). Losing the heavy-weight votes first is exactly why the gradient decays steeply near x.

Vote for $y = x + 2$



The question. Two steps out: does the property generalize to $y = x + 2$?

Who's left. Only the 2 widest intervals still reach y — and those are the *least* likely (thinnest bars). **2 of 7 vote.**

The bar. Few votes, and only low-weight ones → the bar is short.

The payoff. Sweep y across every point and the bar heights trace an **approximately exponential** decay — Shepard's universal law, **derived** from the model, not assumed.

The size principle

Where did the examples come from?

The likelihood $p(X \mid h)$ depends on *how you assume the examples were generated*. Two assumptions:

Weak sampling — examples generated some other way, then *labeled* by whether they fall in h.

Strong sampling — each example drawn *uniformly at random from within* h.

The two likelihoods

Weak sampling

$$p(X \mid h) = \begin{cases} 1 & \text{all } x_i \in h \\ 0 & \text{else} \end{cases}$$

Size-blind: a hypothesis either contains the data or it doesn't. **Does not depend on $|h|$.**

Strong sampling

$$p(x \mid h) = \frac{1}{|h|} \quad \Rightarrow \quad p(X \mid h) = \left(\frac{1}{|h|}\right)^n$$

Each example drawn *uniformly from inside* h. **Smaller** $|h|$ → **higher likelihood**, exponentially so in n.

$|h|$ — the size of hypothesis h (how many stimuli it contains).

n — number of examples.

The size principle

Under strong sampling: **smaller hypotheses get higher likelihood** — and *exponentially* more so as the number of examples n grows.

$$p(X \mid h) = \left(\frac{1}{|h|}\right)^n$$

So a small $|h|$ wins — and **wins faster with more data**, because the exponent n magnifies any size advantage.

The mechanism behind both games today.

Why — the suspicious coincidence

You see the examples $\{60, 80, 10, 30\}$.

- If the concept is **“multiples of 10”** — unremarkable, that’s just what its members look like.
- If the concept is **“even numbers”** — it’s a **suspicious coincidence** that not one of the four was 2, 4, 6, 8, …

Strong sampling *penalizes* the big hypothesis for failing to predict the tight clustering you actually saw.

Poll — strong sampling

What is **strong sampling**?

- **A.** Each stimulus is generated uniformly at random from the true hypothesis
- **B.** A stimulus has probability one given the true hypothesis
- **C.** Larger hypotheses are given smaller prior probability
- **D.** Smaller hypotheses are given smaller prior probability

Poll — answer

A. Uniformly at random from the true hypothesis.

B describes *weak* sampling — membership gives likelihood 1, regardless of size. **C / D** describe a *prior* over hypotheses; the size principle is about the **likelihood**, not the prior. Strong sampling = “the examples were drawn *from inside* the concept” — and that is what makes size matter.

Rectangle game — continuous concepts

Gnarbles

A **gnarble** is a rectangle whose dimensions fall in some interval.

Tenenbaum's cover story: a **2-D concept** is an axis-aligned rectangle in a feature space — e.g. *healthy levels* of insulin × cholesterol.

You observe a few examples drawn from inside the rectangle. **Which other points are gnarbles?**

Start in 1-D

Concept = an **interval** $[\ell, u]$ on one dimension.

- Hypothesis space $\mathcal{H}$ = *all* intervals
- Strong sampling $\rightarrow$ likelihood $\propto \left(\frac{1}{u-\ell}\right)^n$
- Size of the hypothesis = the **length** $u - \ell$

The generalization gradient

Plot $p(y \in C \mid X)$ as y moves along the dimension:

- **High** inside the range of the examples
- **Decays** as y moves outside that range

The decay is *exponential* — Shepard's universal law, now **derived** rather than assumed.

More examples → tighter generalization

- **One example** → broad, diffuse generalization (many interval sizes survive)
- **Many examples** → tight generalization, hugging the data range

Why: the size principle. With large n, big intervals lose likelihood *exponentially* fast — only the small, snug intervals keep posterior mass.

Into 2-D — the rectangle experiment

Same machinery: hypothesis = an axis-aligned **rectangle**.

Tenenbaum's experiment: subjects see dots from "an arbitrary rectangle of healthy levels" and guess the rectangle.

- Measure how far d people extend **beyond** the data range r
- d shrinks as r grows and as the number of dots n grows
- The Bayesian model predicts this d-vs-r-vs-n pattern

One fix — an exponential prior

Pure size principle slightly **over-extends** the rectangle.

The fix: an **exponential prior over rectangle size** — larger rectangles are a priori less likely.

Justified by the task framing (the monitor’s edges, prior experience with rectangles). With it, the model fits the human data.

The rectangle game in one line

The continuous concept learner is **just the framework equation** with $\mathcal{H}$ = intervals / rectangles.

$$p(y \in C \mid X) = \sum_{h} \mathbf{1}[\,y \in h\,]\; p(h \mid X)$$

Nothing new — only the choice of $\mathcal{H}$.

Number game — discrete concepts

The number game

A simple task (Tenenbaum, 1999):

- I have a concept — a set of numbers between 1 and 100
- You see one or more **“yes” examples**
- You judge: is some other number a “yes”?

Watch what your own judgments do as examples come in.

What people actually do

Human generalization judgments (Tenenbaum's $N = 20$ subjects):

- Examples $\{60\}$ → **diffuse** similarity: many numbers get moderate "yes"
- Examples $\{60, 80, 10, 30\}$ → sharp **"multiples of 10"**
- Examples $\{60, 52, 57, 55\}$ → sharp **"numbers near 60"**

One example → graded. Four examples → a crisp **rule**.

Two things to explain

1. Generalization can look **similarity-based (graded)** *or* **rule-based (all-or-none)** — and people switch between them.
2. People learn a concept from **just a few examples**.

One model — the Bayesian framework — produces *both*, with no extra machinery.

The discrete hypothesis space

$\mathcal{H}$ for the number game has two kinds of hypothesis:

Mathematical properties (~24) — even, odd, primes, squares, cubes, multiples of k, powers of k.

Magnitude intervals — "numbers in $[a, b]$": e.g. 10–20, 30–45.

The **prior** $p(h)$ weights these families against each other.

Size principle, by the numbers

Two candidate concepts for numbers in 1–100:

- **Multiples of 2** — 50 numbers → $p(x \mid h) = \dfrac{1}{50} = 2\%$ each
- **Multiples of 10** — 10 numbers → $p(x \mid h) = \dfrac{1}{10} = 10\%$ each

One example: $x = 60$

$$p(60 \mid \text{mult-2}) = \frac{1}{50} \qquad p(60 \mid \text{mult-10}) = \frac{1}{10}$$

Multiples of 10 is already **5×** more likely — but only 5×. With one example, many hypotheses stay in contention $\rightarrow$ **graded** generalization.

Four examples: $\{10, 30, 60, 80\}$

$$p(X \mid \text{mult-2}) = \left(\frac{1}{50}\right)^4 \approx 1.6 \times 10^{-7}$$

$$p(X \mid \text{mult-10}) = \left(\frac{1}{10}\right)^4 = 10^{-4}$$

Now multiples of 10 is **~625×** more likely. The 5× edge got raised to the 4th power → a crisp **rule**.

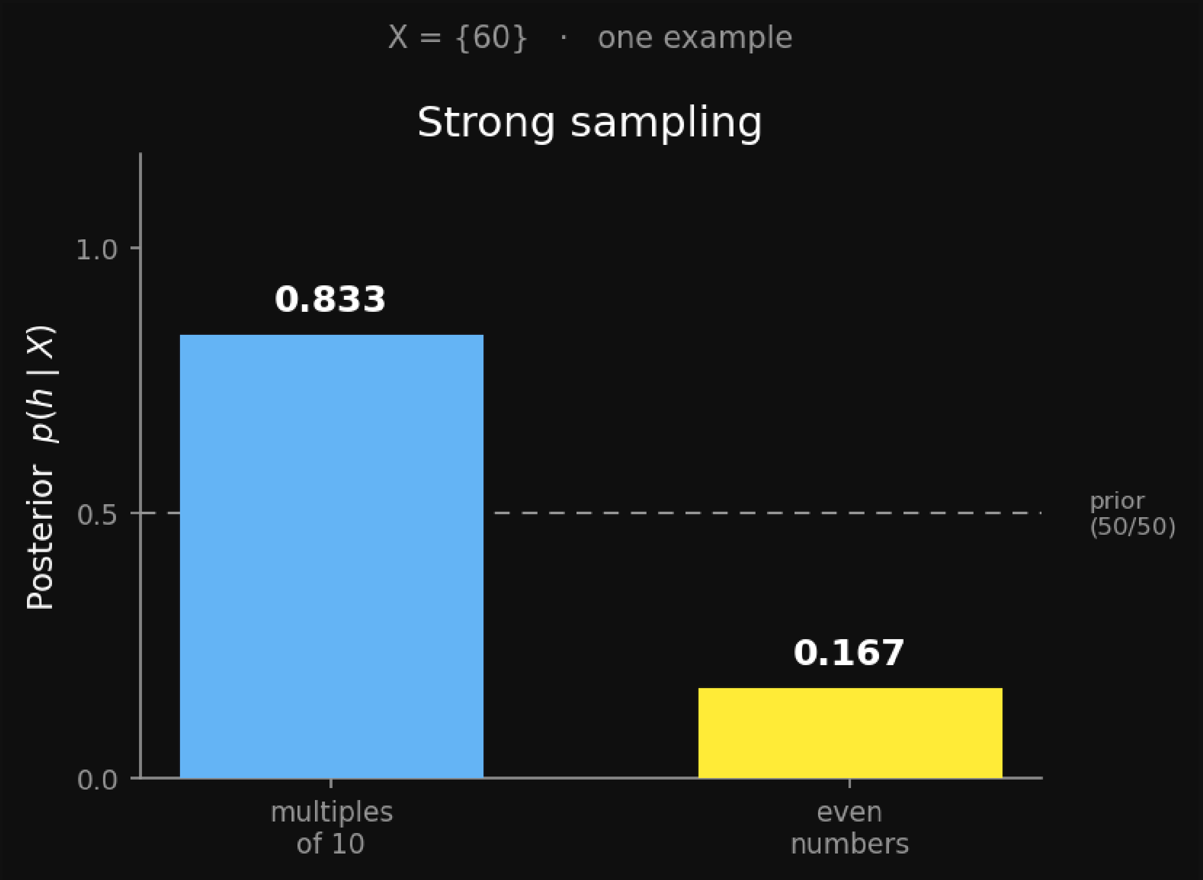
From likelihood to posterior

Likelihoods aren't beliefs yet. Take a **two-hypothesis** model — $\mathcal{H} = \{$ **multiples of 10**, **even numbers** $\}$, both containing every example — with a flat prior, and turn the crank:

$$p(h \mid X) = \frac{p(X \mid h)\, p(h)}{\sum_{h'} p(X \mid h')\, p(h')}$$

Next: the posterior under **strong** vs. **weak** sampling, for $X = \{60\}$ then $X = \{60, 80, 10, 30\}$.

Strong sampling — one example



The data: $X = \{60\}$ — a single example. Strong sampling:

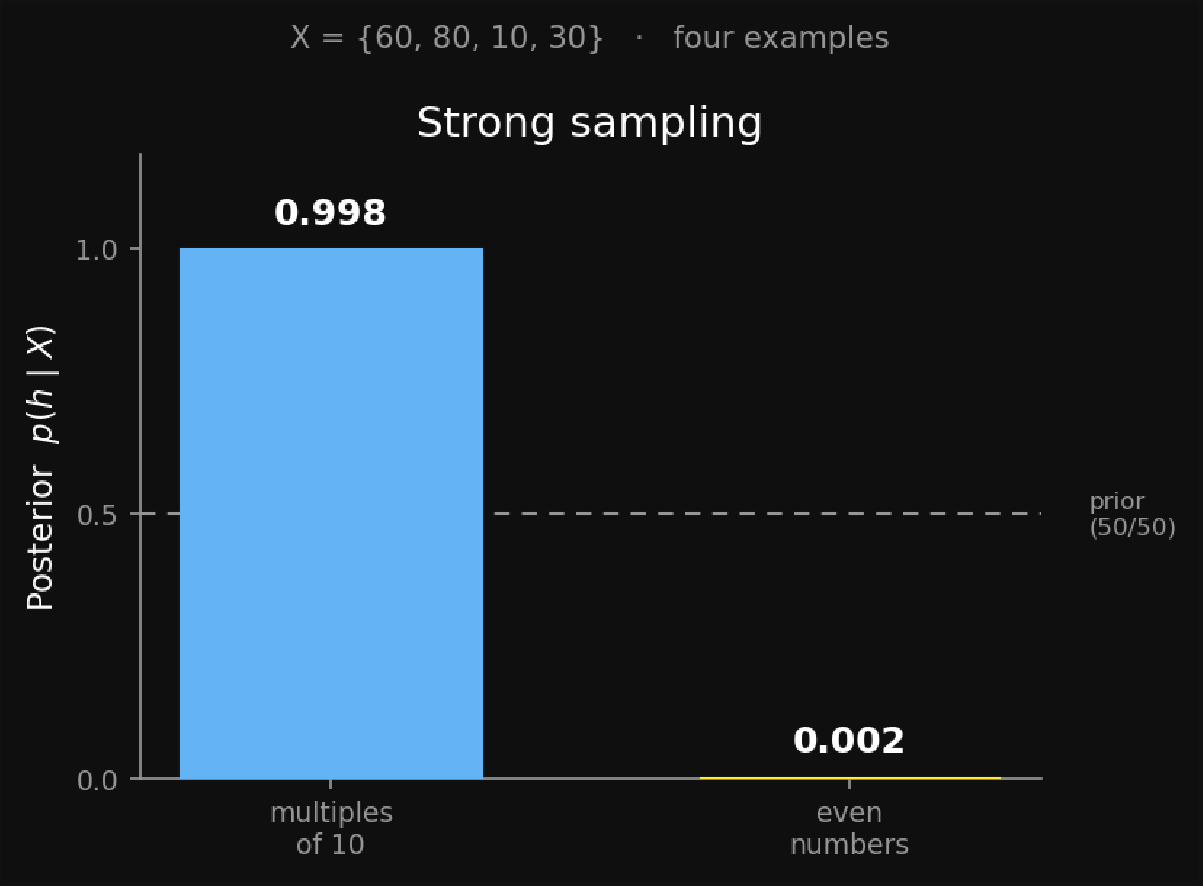
$$p(h \mid X) \propto \left(\frac{1}{|h|}\right)^1 \cdot 0.5$$

- mult-10: likelihood $\frac{1}{10}$ · even: $\frac{1}{50}$
- Normalize the two → **0.83 vs. 0.17**

Read it: one example already **tilts** belief toward the smaller hypothesis — but only gently. A 5:1 likelihood ratio is not decisive, so plenty of posterior mass still sits on “even numbers”.

This is the **graded** regime — belief shifts, but no rule yet.

Strong sampling — four examples



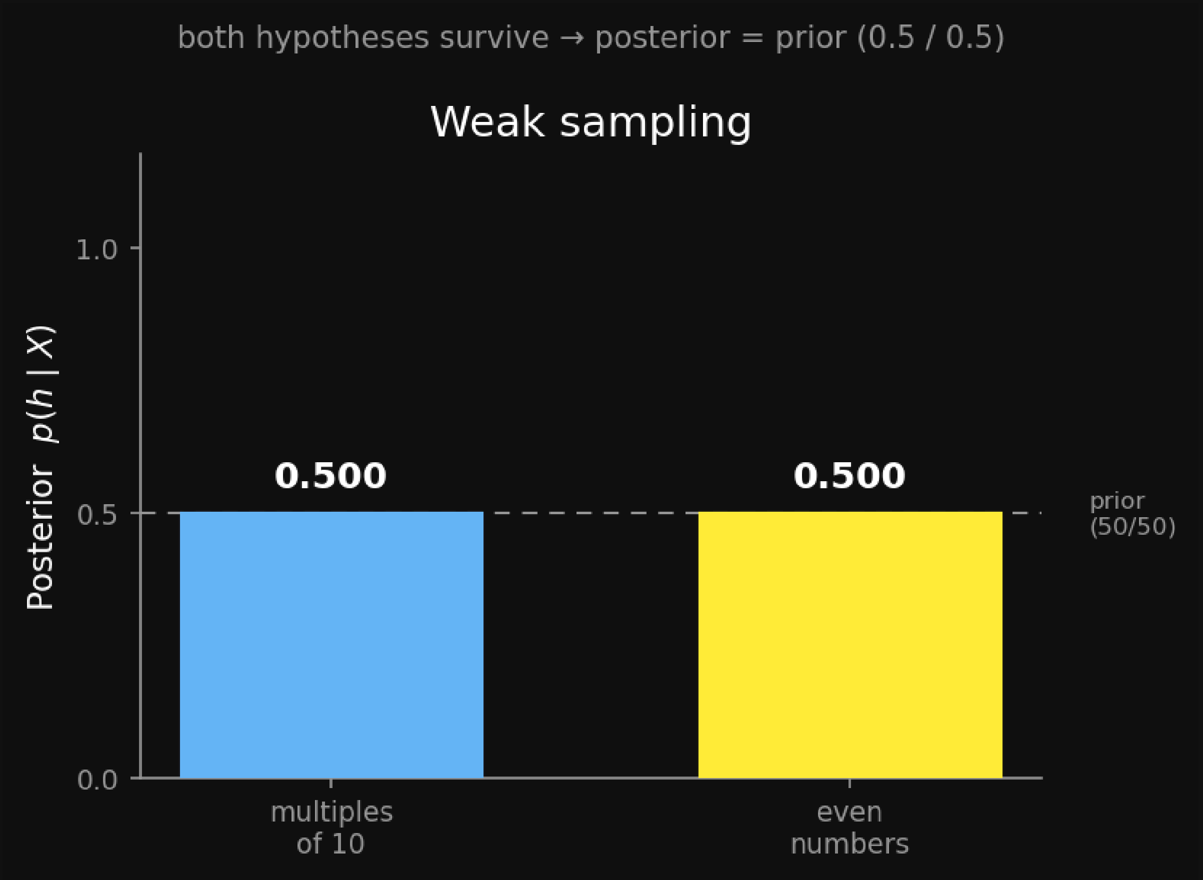
$X = \{60, 80, 10, 30\}$, strong sampling:

$$p(h \mid X) \propto \left(\frac{1}{|h|}\right)^4 \cdot 0.5$$

- The $5{:}1$ ratio is now $5^4 = 625{:}1$
- Normalize → **0.998 vs. 0.002**

Four “even” numbers, none of them odd-looking — that’s the **suspicious coincidence**. Strong sampling all but rules “even” out.

Weak sampling — eliminate, but don't rank



Weak-sampling likelihood is 1 **for any h that contains the data**, 0 otherwise — *no $|h|$ dependence*.

What it can do: a datum *outside* h kills it — likelihood 0. Weak sampling **does** move the posterior, by **ruling hypotheses out**.

What it can't do: among the hypotheses that *all still contain the data*, it has no preference — every survivor keeps likelihood 1, so the posterior over them stays at the **prior**.

Here both hypotheses contain every example — nothing is ruled out:

$$p(h \mid X) = \frac{1 \cdot 0.5}{1 \cdot 0.5 + 1 \cdot 0.5} = 0.5$$

So weak sampling can't see the suspicious coincidence — it can't tell "multiples of 10" from "even numbers".

The number game in one line

Same equation as the rectangle game — $\mathcal{H}$ is now **discrete**.

$$p(y \in C \mid X) = \sum_h \mathbf{1}[\, y \in h \,]\, p(h \mid X)$$

Graded vs. rule-like generalization both **fall out of the posterior** — no extra mechanism. The size principle's exponent does the switching.

Break

Bridge — Shohei's paper

Up next: **Shohei** presents **Tenenbaum & Xu (2000),** ***Word learning as Bayesian inference***.

A child hears *"this is a dax"* pointing at three Dalmatians. Is a poodle a dax? A cat?

This is the **number game**, with $\mathcal{H}$ = candidate word meanings (subordinate / basic-level / superordinate). The **size principle** explains why three subordinate examples → a subordinate meaning.

Watch for the size principle doing the work.

Student presentation — Shohei

No Free Lunch

The No Free Lunch theorem

Wolpert (1996) — averaged over **all possible worlds**, no learning algorithm beats any other.

Sequence prediction: you see x_1, x_2, predict x_3.

- If *every* hypothesis about the sequence is equally possible …
- … then for any data, every prediction is exactly **50/50**.

What NFL means for us

A learner only works because the distribution over worlds is **constrained** — i.e. because it has a **non-flat prior**.

- Generalization is *impossible* without inductive bias
- Recall: the hypothesis space $\mathcal{H}$ and the prior $p(h)$ from Block 3
- They are not bookkeeping — they are *the entire reason* learning is possible

Poll — No Free Lunch

What is the No Free Lunch theorem (for prediction)?

- **A.** When all hypotheses are possible, there's nothing you can learn to predict
- **B.** Learning one hypothesis hurts learning other hypotheses
- **C.** If someone gives you lunch for free, they'll expect something back
- **D.** Generalizing to new stimuli can hurt a learner

Poll — answer

A. When all hypotheses are possible, there's nothing you can learn to predict.

A flat prior over every possible world means data carries no leverage — every continuation stays 50/50. To learn, you must *commit* to some worlds being more likely than others. That commitment is your prior — and it is why your prior matters.

Hierarchical Bayes

Back to Chibany's class

Every student in Chibany's class has their **own** tonkatsu-vs-hamburger rate θ_i.

The rates aren't identical — but they aren't unrelated either. They're all Chibany's customers.

Should learning Aoi's rate tell us anything about Ben's?

Priors over priors

Put a prior on the **parameters of the prior** — and treat (a, b) themselves as unknown:

$$\theta_i \sim \mathrm{Beta}(a, b)$$

The two-level model: $(a, b) \;\longrightarrow\; \theta_i \;\longrightarrow\;$ observed bentos.

(a, b) is the *shared* structure; each θ_i is a student.

Why it matters

A hierarchical model lets a learner **learn the prior from data** — exactly the inductive bias No Free Lunch said you cannot do without.

- (a, b) is learned from *all* the students together
- It is how “overhypotheses” get acquired — the shape bias, object vs. substance kinds (Kemp, Perfors & Tenenbaum, 2007)

Three ways to pool the data

Six students, six bento records. How do you estimate their rates?

Approach	Model	Problem
Complete pooling	one shared θ for everyone	ignores that students differ
No pooling	a separate θ_i, all unrelated	ignores that they're all Chibany's customers
Hierarchical	$\theta_i \sim \mathrm{Beta}(a, b)$	*the middle path* — borrows strength

Hierarchical Bayes = **partial pooling**.

The two-level model

Built up, top to bottom:

$$(a, b) \ \sim \ \text{prior}$$

$$\theta_i \mid a, b \ \sim \ \text{Beta}(a, b)$$

$$k_i \mid \theta_i \ \sim \ \text{Binomial}(n_i, \theta_i)$$

k_i — tonkatsu count for student i. n_i — that student's total bentos.

Inference — no closed form

We want the posterior over *everything*:

$$p\big(a, b, \{\theta_i\} \mid \mathrm{data}\big)$$

Unlike the Beta-Binomial of Week 3, this has **no clean closed form**.

This is where sampling comes in — and where **GenJAX** earns its place (the hierarchical `bento_day()` exercise builds exactly this model).

Shrinkage — borrowing strength

For each student, compare:

- their **raw** tonkatsu fraction k_i/n_i
- their **posterior mean** $\hat{\theta}_i$

The posterior means are pulled **toward the group mean** — *a lot* when a student has little data, barely at all when they have a lot.

This is hierarchical Bayes **automatically borrowing strength** across students.

Overhypotheses — learning the bias

With the hierarchy in hand, **overhypotheses** become precise.

The shape bias; the object-vs-substance distinction — these are *second-level* hypotheses, learned as a distribution over *kinds of concept* (Kemp, Perfors & Tenenbaum, 2007).

No Free Lunch said a learner needs inductive bias. The hierarchy is where a learner **acquires** it — instead of being born with it.

Close

Before next week

Read **T3 Ch 5 — Mixture models** before Week 5.

It formalizes the same partial-pooling idea you just saw, and feeds directly into **Problem 3** of the Clusters assignment.

Clusters is due **Fri Jun 5, 8:00 PM**.

Next week — Week 5

Bayes nets + causal Bayes nets.

From “concepts as sets” to **structured probabilistic models** — graphs that encode how variables depend on each other, and what happens when you *intervene*.

Check the readings page for the Week 5 paper + presenter.